

PROJECT SYNOPSIS

MeterFlow

A Multi-Tenant SaaS Metered Billing Platform with Predictive Spend Alerts

Project Title	MeterFlow
Project Domain	Software Engineering / Artificial Intelligence
Core Areas	Multi-Tenant Architecture, Metered Billing, Predictive Analytics, Secure Authentication, Distributed Systems
Proposed Platform	Web-based SaaS application

1. Introduction

Software-as-a-Service (SaaS) has become one of the most widely adopted software delivery models, enabling organizations to access cloud-based applications without maintaining dedicated infrastructure. Many SaaS platforms use subscription-based pricing models where customers are charged based on the resources they consume, such as API requests, storage, or active users. Managing this usage accurately while ensuring data isolation between multiple organizations is a significant challenge.

MeterFlow is a Multi-Tenant SaaS Metered Billing Platform designed to address these challenges by providing a centralized system for tracking resource usage, calculating usage-based bills, and generating predictive spend alerts. The platform supports multiple organizations on a shared application while maintaining secure separation of tenant data. In addition, it incorporates machine learning techniques to analyze historical usage patterns and estimate future spending, enabling organizations to monitor costs proactively.

The project demonstrates concepts such as multi-tenant architecture, metered billing, predictive analytics, secure authentication, and distributed locking, making it a comprehensive solution for modern cloud-based applications.

2. Problem Statement

Many organizations use SaaS platforms that charge customers based on resource consumption. However, traditional billing systems often provide only historical billing information and lack intelligent forecasting capabilities. They also face challenges in maintaining accurate billing records when multiple organizations share the same application, especially during concurrent billing operations.

There is a need for a secure and scalable platform that can manage multiple tenants, accurately track resource usage, generate metered bills, and provide predictive spending insights to help organizations make informed financial decisions.

3. Objectives

- Design and develop a multi-tenant SaaS platform capable of supporting multiple organizations.
- Track resource usage independently for each tenant.
- Implement accurate usage-based billing using high-precision financial calculations.
- Generate invoices automatically based on resource consumption.
- Predict future spending using historical usage data and machine learning techniques.

- Display billing analytics and spending trends through an interactive dashboard.
- Prevent duplicate billing operations using Redis-based distributed locking.
- Develop a scalable and modular architecture following modern software engineering practices.

4. Scope

The project focuses on developing a web-based SaaS billing platform for organizations that operate on a usage-based pricing model. The platform will provide secure authentication, tenant management, usage tracking, invoice generation, billing analytics, and predictive spend alerts.

The project will demonstrate the complete billing workflow from resource usage tracking to invoice generation and expenditure prediction. The machine learning component will forecast future spending using historical billing data.

The project is intended for educational purposes and will simulate billing operations using generated datasets. Real payment gateway integration and production-scale cloud deployment are outside the scope of this implementation.

5. Proposed Solution / Methodology

The proposed system follows a modular client-server architecture consisting of a React frontend, FastAPI backend, PostgreSQL database, Redis cache, and a machine learning module.

Workflow

- 1 Users register and log into the platform.
- 2 An organization (tenant) is created.
- 3 Users are assigned to their respective organizations.
- 4 Resource usage is continuously recorded.
- 5 The billing engine calculates charges based on predefined pricing rules.
- 6 Invoices are generated automatically.
- 7 Historical billing data is stored in the database.
- 8 The machine learning model analyzes previous spending patterns.
- 9 Future expenditure is predicted using Linear Regression.
- 10 The dashboard displays billing information, analytics, and predictive spend alerts.

Implementation Approach

- Multi-tenant architecture using tenant-specific data isolation.
- RESTful APIs for communication between the frontend and backend.
- PostgreSQL for persistent data storage.
- Redis distributed locking to prevent concurrent billing conflicts.
- Linear Regression model for spend prediction.
- Interactive dashboard for monitoring usage and financial analytics.

6. Features / Modules

Module	Features
1. User Authentication	User Registration; Login; Secure Authentication; Password Encryption
2. Organization Management	Create Organization; Manage Tenant Information; User Management
3. Subscription Management	Subscription Plans; Plan Assignment; Subscription Status
4. Usage Tracking	API Usage Monitoring; Storage Usage; Resource Consumption Tracking
5. Billing Module	Metered Billing; Invoice Generation; Billing History; Financial Reports
6. Predictive Analytics	Historical Data Analysis; Spend Forecasting; Predictive Spend Alerts
7. Dashboard	Usage Analytics; Revenue Summary; Billing Statistics; Spending Trends
8. Administration	Organization Management; User Monitoring; Billing Configuration

7. Technology Stack

Category	Technology
Programming Languages	Python; TypeScript; SQL
Frontend	React; Tailwind CSS
Backend	FastAPI
Database	PostgreSQL
Machine Learning	Scikit-learn; Pandas; NumPy
Caching & Concurrency	Redis
Development Tools	Git; GitHub; Visual Studio Code; Postman
Deployment	Docker

8. System Requirements

Hardware Requirements

Minimum	Recommended
Intel Core i3 Processor	Intel Core i5 / AMD Ryzen 5 or above
8 GB RAM	16 GB RAM
256 GB Storage	SSD Storage

Software Requirements

- Windows 10/11, Linux, or macOS
- Python 3.12+
- Node.js
- PostgreSQL
- Redis
- Git
- Visual Studio Code
- Docker Desktop (Optional)

Browser Requirements

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

9. Expected Outcomes

- Provide a secure multi-tenant SaaS platform.
- Track organizational resource usage accurately.
- Generate automated usage-based invoices.
- Display real-time billing analytics through an interactive dashboard.
- Predict future spending using machine learning.
- Help organizations monitor and optimize operational costs.
- Demonstrate the practical implementation of SaaS architecture, billing systems, and predictive analytics.

10. Future Scope

- Payment Gateway Integration (Stripe/Razorpay)
- Email Invoice Delivery
- AI-powered Cost Optimization Recommendations
- Role-Based Access Control (RBAC)
- Multi-Currency Billing
- Real-Time Notifications
- Kubernetes-based Cloud Deployment
- Advanced Time-Series Forecasting Models (ARIMA, Prophet, LSTM)
- Mobile Application
- Audit Logs and Compliance Reports

11. Conclusion

MeterFlow aims to provide a complete SaaS metered billing solution by combining multi-tenant architecture, usage-based billing, predictive analytics, and secure backend design into a single platform. The project demonstrates the practical implementation of modern cloud application concepts while addressing real-world challenges related to billing accuracy, resource monitoring, and financial forecasting.

By integrating software engineering principles with machine learning techniques, the platform offers organizations valuable insights into their resource consumption and future expenditure, making it a comprehensive academic project with practical industry relevance.